Differentiation

- Differentiability: $f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$ exists
- Smooth: $f^{(k)}$ exists and is continuous
- C^k: $f^{(k)}$ exists and is continuous
- Smooth: C^∞ $\forall k \in \mathbb{N}$
- Gradient: $\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right)$
- Directional derivative in direction v: $\nabla f(x) \cdot v$
  - instantaneous rate of change, decide by $\|v\|$
- Direction of greatest increase of f
  - Pf: Let θ be the angle between $\nabla f(x)$ and v. Then $\nabla f(x) \cdot v = \|\nabla f(x)\| \|v\| \cos\theta$
  - orthogonal to level curve $x(t)$ (such that $f(x(t)) = c$)
  - Pf: $\frac{d}{dt} f(x(t)) = \frac{df(x(t))}{dt} = \nabla f(x) \cdot \frac{dx}{dt} = 0$
- Jacobian: $f: \mathbb{R}^n \to \mathbb{R}^m$ $\to J_f(x) = (\nabla f_1(x), \dots, \nabla f_m(x))^T$
  - Jacobian determinant tells how much the volume changed by f
  - $\det(J_f(x_0)) \neq 0 \Rightarrow f$ invertible in neighborhood of x_0
- Hessian: $H_f(x) = J_{\nabla f}(x) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)_{ij} = J_{J_f(x)}$
  - Equivalent of Jacobian of gradient
  - Equivalent of Jacobian of gradient
  - Equivalent of Jacobian of gradient
- Lipschitz-continuity: $\|f(x) - f(y)\| \leq L\|x - y\|$
  - slopes are bounded: $\|f(x) - f(y)\| \leq L\|x - y\|$
  - diff. conv. and f differentiable $\Rightarrow f$ Lipschitz continuous
- Lipschitz Gradients: Lipschitz on Jacobian of ∇f (Hessian)
  - Equivalent (curvature) of ∇f are bounded by L
- multivariate differentiation rules:
  - $\frac{d}{dx}(f+g) = \frac{df}{dx} + \frac{dg}{dx}$
  - $\frac{d}{dx}(fg) = \frac{df}{dx} g + f \frac{dg}{dx}$
  - $\frac{d}{dx}(f \circ g) = \frac{df}{dg} \frac{dg}{dx}$
  - $\frac{d}{dx}(f^{-1} \circ f) = \frac{df^{-1}}{dx} \frac{df}{dx} = 1$
  - $\frac{d}{dx}(f^{-1} \circ f) = \frac{df^{-1}}{df} \frac{df}{dx}$
- helpful:
  - $\frac{d\Sigma x_i}{dx_k} = \frac{d\Sigma^T}{dx_k} x^T = \frac{d\Sigma x}{dx_k} = \frac{d\Sigma x}{dx} = \Sigma^T$
  - $\frac{d\Sigma x^T x}{dx_k} = \frac{d\Sigma x^T}{dx_k} x + \frac{d\Sigma x}{dx_k} x = \frac{d\Sigma x^T x}{dx} = \Sigma^T (x + x^T)$
  - $\frac{d}{dx} f(x)^T = \frac{d}{dx} f(x)^T = \frac{d}{dx} f(x)^T = \frac{d}{dx} f(x)^T = \frac{d}{dx} f(x)^T$
  - $\frac{d}{dx}(a^T x^T x) = \frac{d}{dx} a^T (x^T x)^T$
  - $\frac{d}{dx} \frac{y^T x}{x^T x} = y^T x \frac{d}{dx} \frac{1}{x^T x} + \frac{d}{dx} y^T x \frac{1}{x^T x}$; $y, x: \mathbb{R}^n \to \mathbb{R}$
  - $\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{g}{g^2} \frac{df}{dx} - \frac{f}{g^2} \frac{dg}{dx}$
  - $\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{1}{g} \frac{df}{dx} - \frac{f}{g^2} \frac{dg}{dx}$
- Example: Logistic regression negative log likelihood
  - $-\ell(\theta) = \sum_{i=1}^n -y^{(i)} \log[\sigma(x^{(i)}\theta)] - (1-y^{(i)}) \log[1 - \sigma(x^{(i)}\theta)]$
  - $\log x^{(i)}(\theta) = s(\theta^T x) = \frac{1}{1+e^{-\theta^T x}}$
  - $\Rightarrow \nabla_\theta - \ell(\theta) = -\sum_{i=1}^n \frac{\partial}{\partial \theta} \frac{\partial \ell}{\partial \theta} = -\sum_{i=1}^n (s(\theta^T x) - y^{(i)}) x^{(i)}$

Convexity
- Convex Set:
  - $\forall x, y \in S$ and $\forall \theta \in [0,1]$: $\theta x + (1-\theta) y \in S$
- Convex Function:
  - $f: S \to \mathbb{R}$ convex $\Leftrightarrow \forall x \neq y \in S$ $\forall \theta \in (0,1)$: $f(\theta x + (1-\theta) y) \leq \theta f(x) + (1-\theta) f(y)$
  - interpreting line segments from $(x, f(x))$ to $(y, f(y))$
  - lies above the line segment
  - Jensen's inequality: $f(\mathbb{E}[x]) \leq \mathbb{E}[f(x)]$ $\forall$ convex f
- Operations preserving Convexity:
  - weighted sum with $w \geq 0$: f convex $\Rightarrow \sum w_i f$ convex
  - g convex, f linear $\Rightarrow g \circ f$ convex
  - elementwise max: f convex $\Rightarrow g(x) = \max_i \{f_i(x)\}$ convex
  - f convex and non decreasing, g convex $\Rightarrow f(g(x))$ convex
  - h function, f convex, g linear $\Rightarrow$ $f(h(x))$ convex

Second order condition: f convex $\Leftrightarrow \nabla^2 f(x) \succeq 0$ (pos. semi definite)

Quadratic Functions

- $q(x) = x^T A x + b^T x + c$
- Convexity/concavity depend on Eigenvalues of $H = 2A$
  - all Eigenvalues $\geq 0 \Rightarrow$ convex, > 0 strict convex
  - $\exists$ pos and negative Eigenvalues $\Rightarrow$ saddle point
- direction of highest curvature is Eigenvector of A max with corresponding λ_{max}
  - $\nabla^2 q(x) = \nabla(\nabla q(x)) = \nabla(Ax + b) = \frac{1}{2}(A + A^T) = A$ sym
  - condition: $\|\nabla q(x)\| \geq \epsilon \|x\|$
  - easy if right condition but for optimization (unstable)

Unconstrained Optimization

- Optimizing $f: S \to \mathbb{R}$ - unconstrained if $S = \mathbb{R}^d$
  - unconstrained if $\dim S$ glob. minimum
  - strict if f at least $\leq f(x^*)$
  - convex if at least $f(x^*)$
- Maximum Likelihood Estimation: $\hat{\theta}_{MLE} = \arg\max_\theta \prod_{i=1}^n p_\theta(x^{(i)})$
  - minimizes Loss of normal data: $\hat{\theta}(x_i, y_i) = \arg\min_\theta \sum_{i=1}^n \frac{1}{2\sigma^2} (y_i - \theta^T x_i)^2$
  - $\Rightarrow \sum_{i=1}^n (y_i - \theta^T x_i)^2 = \|y - X\theta\|_2^2$
  - $\Rightarrow \hat{\theta}_{MLE} = (X^T X)^{-1} X^T y$
- $\|R_{reg}\|$ Risk Min: $\min_\theta \sum_{i=1}^n L(y^{(i)}, f_\theta(x^{(i)})) + \lambda \|\theta\|$
  - consists of:
    - Hyp Space parameter space
    - Risk measure prediction error
    - Regularization penalty model complexity
    - Optimization: minimize Risk - pseudo goal
  - Ridge Reg: convex & smooth
  - Lasso Reg: convex but not smooth
  - $\|\theta\|_1$ not differentiable $\Rightarrow$ Lipschitz continuous
- Lipschitz Gradients: Lipschitz on Jacobian of ∇f (Hessian)
  - Equivalent (curvature) of ∇f are bounded by L
- multivariate differentiation rules:
  - $\frac{d}{dx}(f+g) = \frac{df}{dx} + \frac{dg}{dx}$
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  - $\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{g}{g^2} \frac{df}{dx} - \frac{f}{g^2} \frac{dg}{dx}$
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  - $\log x^{(i)}(\theta) = s(\theta^T x) = \frac{1}{1+e^{-\theta^T x}}$
  - $\Rightarrow \nabla_\theta - \ell(\theta) = -\sum_{i=1}^n \frac{\partial}{\partial \theta} \frac{\partial \ell}{\partial \theta} = -\sum_{i=1}^n (s(\theta^T x) - y^{(i)}) x^{(i)}$

Convexity

Evolutionary algorithm - Numerical Encodings

Recombination
- Uniform Crossover: $y_{new}[i] = y_1[i]$ with probability p and of parent 2 with probability $1-p$
- Intermediate recombination: Offspring as mean of parents
- Line Recombination: generate α offspring:
  $\Rightarrow \frac{1}{2}(x+x) \pm \frac{1}{2}\beta(x-x)$; $\beta \in [0,\alpha]$

Mutation
- Uniform: $x_{new} \sim U(x-a, x+a)$
- Gauss: $x_{new} \sim N(x, \sigma)$
- any other distribution (Polynomial!)

GA / Bit Strings

Binary Encoding: typically for problems with binary decision variables (e.g. feature selection)

Recombination
- 1-point crossover: first k bits from parent 1, last dim from parent 2
- Uniform crossover: Select bit y_i with probability p from parent 1, else from parent 2

Mutation: Flip of each bit with probability p
- doesn't make sense if bits are binary numbers
- e.g. Feature selection (1 means feature included)

CMA-ES Algorithm

→ Estimation of Distribution
- Draw λ offspring x_i from $p(f|\xi_k)$
- Evaluate fitness $f(x_i)$
- Update ξ_{k+1} as best offspring
- move distribution towards optimum

Covariance Matrix adaptation
- $x_i \sim m_k + \sigma_k N(0, C_k)$; (i.th offspring at time t)
- m_{k+1} = weighted average of best x_i ; $\{i \in \{1,\dots,\lambda\}\}$
- C_k : covariance matrix of best x_i ; $\{i \in \{1,\dots,\lambda\}\}$
- σ_k : overall st dev (→ step size)

Steps: 0) Initialize $m_0, \sigma_0, C_0 = I_d$
- 1) Sample size λ: $x_i = m_k + \sigma_k N(0, C_k)$
- 2) Select the best performing x_i
- 3) Set weights w_i with $\sum w_i = 1$ where $x_i^{(i)}$ is i-th best performing x ($w_1 \geq w_2 \geq \dots$)
- 4) Calculate $y_w^{(i)} = \sum_{i=1}^{\lambda} w_i(x_i^{(i)} - m_k)$; $\sigma_k = \sum_{i=1}^{\lambda} w_i x_i^{(i)}$
  step size σ might be given
- 5) Set $m_{k+1} = m_k + \sigma_k \cdot y_w$
- 6) Calculate $C_{k+1} = (1-\lambda)C_k + \frac{1}{\sigma^2}\sum_{i=1}^{\lambda} w_i (x_i - m_k)(x_i - m_k)^T$
- updating σ_k with mutation and selecting best

Bayesian Optimization

Black Box: $f: S \to \mathbb{R}$; S: box constrained
- no gradient of f → can't use Adam or no analytic description
- expensive evaluation of f → no standard derivative-free methods

Naive Approaches
- Grid Search: never good → lots of information lost if high correlation or in variable having impact
- Random Search: ok for few dimensions
  ↳ But: inefficient, bc. search in irrelevant areas
- Nelder-Mead, simulated annealing, EA: inefficient

Basic BO Loop & Surrogate Modelling

Surrogate Modelling → approximate f
- f unknown and expensive

→ Initial Design (evaluations at beginning)
- be not too small or too large → at least $d+1$
- Latin Hypercube Sampling
  ↳ partition each axis into k of equal probabilities (discretize input)
  ↳ for each axis exactly one sample per bin
- Maximin sampling:
  ↳ minimal distance between any two points is maximized (e.g. choose q is packing radius)
  ↳ find D that maximizes $\min_{x<x'} \|x-x'\|$
  ↳ most exploration

→ Fit surrogate model $\hat{f}$ on $D_t = \{(x_i, y_i)\}_{i=1,\dots,t}$
→ Find min of $\hat{f}$ (continuous optimization problem)
→ evaluate $\hat{f}$ at that point to get D_{t+1}

→ But: high exploitation, low exploration

Posterior Uncertainty & Acquisition Functions

Probability of Improvement
→ Goal: find x_{pos} that maximizes Probab. of Improvement
- $a_{PI} = P(f(x) < f_{min}) = \Phi\left(\frac{f_{min} - \hat{\mu}(x)}{\hat{\sigma}(x)}\right)$
  ↳ if posterior mean, $\hat{\sigma}$ posterior st. dev.
- because $\hat{\sigma}$ expected value is further away, with higher variance, more probability mass could fall below f_{min}
  ↳ $a_{PI}(x) = 0$ only for design points (already known) because $\hat{\sigma}(x) = 0$ and $\phi(-\infty) = 0$
  → But does not take size of improvement into account → often propose points close to current x^*

Expected Improvement
- $a_{EI}(x) = E(\max\{f_{min} - Y(x), 0\})$; $Y_{pos} \sim Y(x) = N(\hat{\mu}, \hat{\sigma}^2)$
  ↳ $E(Y(x))$ doesn't make sense because $E(Y(x)) = \hat{\mu}_{pos} - E(Y(x)) = \hat{\mu}_{pos} - \hat{\mu}(x)$ (just function mean)
  ↳ $= E(\max\{E(Y(x)), 0\})$ vanishes high variance
  ↳ $a_{EI}(x) = E(\max\{\hat{\mu}_{pos} - Y(x), 0\})$
  $= \hat{\sigma}_{pos} \cdot \hat{\sigma}(x) \cdot \phi\left(\frac{f_{min} - \hat{\mu}(x)}{\hat{\sigma}(x)}\right) + \hat{\sigma}(x) \cdot \Phi\left(\frac{f_{min} - \hat{\mu}(x)}{\hat{\sigma}(x)}\right)$
  ↳ if $Y(x) \sim N(\hat{\mu}(x), \hat{\sigma}^2(x))$
  ↳ $a_{EI}(x) = 0$ at design points
  → Pro: global Optimizer

Surrogate Models

Gaussian Process
- Posterior Predictive distribution: for test point x: $Y(x) | x, D_t \sim N(\hat{\mu}(x), \hat{\sigma}^2(x))$
  ↳ $\hat{\mu}(x) = k(x,X) K^{-1} y$
  ↳ $\hat{\sigma}^2(x) = k(x,x) - k(x,X)^T K^{-1} k(x,X)$
  ↳ $(y = (y_1, \dots, y_t)$; $k(x,X) = (k(x,x_1), \dots, k(x,x_t))^T$
  ↳ Gauss kernel: $k(x,x') = \exp\left(-\frac{\|x-x'\|^2}{2l^2}\right)$

Pro
- smooth
- good uncertainty estimates
- posterior predictive is a gaussian

Con
- scales cubically in number of data points
- only numeric inputs
- numerical problems might occur

Random Forest
- multiply decision trees on bootstrap sample with feature subsample $\frac{d}{3}$
- mean prediction: $\hat{f}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x)$
- uncertainty proxy: $\hat{\sigma}^2(x) = \left[\frac{1}{B}\sum_{b=1}^{B} \hat{\sigma}_b^2(x) + \hat{f}_b(x)^2\right] - \hat{f}(x)^2$
  → with frequentist split points in trees, the surrogate function is a step function
  Con: weak uncertainty estimates
  → no real posterior predictive (not really Bayesian)
  Pro: cheap to train and scales well
  - robust

Multi-criteria Optimization

Def: $\min_{x \in S} \hat{=} \min\{f_1(x), \dots, f_m(x)\}$, $S \subseteq \mathbb{R}^n$
- no minimizer $f_1, \dots, f_m$ simultaneously
- x_1 Pareto-dominates x_2, if
  ↳ $f_1(x_1) \leq f_1(x_2)$ und $\exists i: f_i(x_1) < f_i(x_2)$
  ↳ $f(x_1) < f(x_2)$
- x^* is pareto optimal if not dominated by any x
- All Pareto optimal candidates: Pareto Set
  $\mathcal{P} = \{x \in S \mid \nexists x' \in S: f(x') \prec f(x)\}$
- Pareto Front: $\mathcal{F} = \{f(x)\}$

→ most either-or-choice, we can't choose best solution
→ a-priori approach: include preference from start
→ a-posteriori approach: choose from Pareto set
A-priori
- create one-dim function via weights:
  $\min_{x \in S} \sum_{i=1}^{m} w_i f_i(x)$
  ↳ Utilitarian
  If $w \geq 0$: $\forall i: x^* = \arg\min_{x \in S} \sum_{i=1}^{m} w_i f_i$ is Pareto optimal
  ↳ Proof: Suppose x^* not optimal
  $\Rightarrow \exists x: \forall i: f_i(x) \leq f_i(x^*) \vee \exists i: f_i(x) < f_i(x^*)$
  $\Rightarrow \exists w: \sum_{i=1}^{m} w_i f_i(x) < \sum_{i=1}^{m} w_i f_i(x^*)$
  Other way around doesn't work:
  Theorem: properly Pareto-optimal
  ↳ we can only find convex pareto front with $x^* = \arg\min_{x \in S} \sum_{i=1}^{m} w_i f_i(x)$
  ↳ Solution x^* is properly Pareto optimal
  ↳ Excludes solution where small improvement in one dimension lead to large decrease in other
  ✓ Properly Pareto optimal if
  $\exists M > 0: \forall i: \forall x: f_i(x) < f_i(x^*)$
  $\Rightarrow \exists i: \exists j: f_j(x^*) < f_j(x)$ and $\frac{f_i(x^*) - f_i(x)}{f_j(x) - f_j(x^*)} \leq M$
  ↳ Lexicographic method
  - minimize f_1 in order
  - ↳ under all possible optimal x for f_1, find the optimal one for f_2
  - ↳ different orders give different solutions

A-Posteriori and Evolutionary Algorithms

Generate some time to estimate Pareto-Front
- only thing on y-space is survival selection

pseudo code: 1) initialize population $P_0 \in S$, $|P_0| = \mu$
- 2) Select from P_t and generate offspring Q_t , $|Q_t| = \lambda$
- 3) Select to survivors for P_{t+1}
- 4) Step -if criterion fulfilled

Selection Algorithm
- NSGA-II
  - (μ+λ) strategy
  - any standard variation algorithm
  - selection: → main: non-dominated sorting (layers)
    → secondary: Crowding distance
  - ↳ non-dominated sorting
    - select all points on Pareto front, then remove, find all points on new Pareto-front, ...
  - ↳ Crowding distance (for partially included front)
    - ↳ prefer more spread-out solutions
    - ↳ find in each objective for each point distance to next neighbour
    - ↳ average over all objectives
- SMS (S-Metric Selection)
  - exclude points with least hypervolume contribution

pseudo code: - generate P_0 of size μ
- create offspring (mutation, recombo)
- non-dominated sorting (not least F_n)
- exclude points with minimal Hypervolume contribution

Bayesian Optimization

pro: efficient sampling, handles noise
- no gradients required

code: (least feasible f - most $\hat{f}$)
- acquisition function (a: max evals T)
- initialize D_0 with initial observations
- for t in D_{t-1}
  - $x_t = \arg\max_x u(x, D_{t-1}, \hat{f}_t)$
  - add $x_t, f(x_t)$ to D_t
- Small-cut: adapt to multiple cost functions
  a) Simplify to 1-dim cost function
  $\min_x \sum_i w_i f_i$; $w_i \geq 0$ → How set w_i?
  ↳ or Pareto EGO (Efficient Global Optimization)
  ↳ with Tchebycheff Norm:
  $\hat{f} = \max_i (w_i \hat{f}_i(x)) + \rho \sum_i w_i \hat{f}_i(x)$
  → weight drawn from $w_i = \frac{k}{S}, k \in \{0, \dots, S\}$ in each iteration

Pareto EGO code:
- initialize D_0 with initial observations
- Sample w with $w_i = \frac{k}{S}, k \in \{0, \dots, S\}$
- $\xi_t = \max(w_i \hat{f}_i(x)) + \rho \sum w_i \hat{f}_i(x)$
- for t in D_{t-1}
  - select $x_t = \arg\max_x u(x, D_{t-1}, \hat{f}_t)$
  - add $(x_t, f(x_t))$ to data

Hypervolume based acquisition functions
- Define acquisition function that models the Hypervolume contribution of new points
  → one acquisition function for each objective
- Problem: large regions where HV add is 0

Code: - initialize D_0 with initial observations
- for t in D_{t-1} ... find on D_{t-1}
  - select $x_t = \arg\max u(x, D_{t-1}, \hat{f}_{t,1}, \dots, \hat{f}_{t,m})$
  - add $(x_t, f(x_t))$ to D_t

$Var(X) = E(X^2) - E(X)^2$

$Var(aX+b) = a^2 Var(X)$

$Var(X+Y) = Var(X) + Var(Y) + 2 \cdot Cov(X,Y)$

→ Quadratic in w ; $Var(X)$ is p.s.d.
⇒ convex

Ridge Regression: $L(\theta | Y, X) = \|Y - Xw\|^2 + \lambda \|\theta\|^2$

Normal distr: $p(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-\mu)^2}$; $\Sigma^{-1}(y-\mu)$

Log.st. Reg: $P(y=y_1 | x_1) = P(y=1|x)^y \cdot P(y=0|x)^{1-y}$
↳ $P(y=y_1|x_1) = \pi(x, \theta) = \frac{1}{1+e^{-\theta^T x}}$
↳ $\ell(\theta|x,y) = \sum_{i=1}^{n} y_i \cdot \log(\pi(x_i, \theta)) + (1-y_i) \cdot \log(1-\pi(x_i, \theta))$
↳ $R_{emp}(\theta) = -\log \sum_{i=1}^{n} P(y=y_i | x_i)$
$= -\sum_{i=1}^{n} y_i \theta^T x_i - \log\left(1 + \exp(\theta^T x_i)\right)$
↳ $\frac{\partial^2}{\partial \theta^2} R_{emp}(\theta) = \frac{\sigma(\theta^T x)}{(1+e^{-\theta^T x})^2} > 0$
↳ when perfect separation:
$x_i^T \beta > 0 \Rightarrow P(y=1 | x_i^T \beta \leq 0$ for $y_i = 0$

$E_{x,y}(\dots) = E_x(E_{y|x}(\dots))$

Pareto Front visualisieren:
- p_{θ_1} (sich richtig $\frac{1}{2}$)

Lagrange - Funktionen
1) zu minimierende Gleichung $f(x,y)$
- Nebenbedingung $g(x,y) = 0$
→ Lagrange-Funktion: $L(x,y,\lambda) = f(x,y) + \lambda \cdot g(x,y)$
2) Ableitung nach x
Ableitung nach y
Ableitung nach λ
$\Big\} = 0$ setzen
3) Gleichungssystem lösen

For which constant α does GD converge for $f(x) = x^T A x + c$
- ↳ A symmetric p.d. ⇒ $A = V \Lambda V^T$
- ↳ define $w = V^T x$ ⇒ $x = V w$
- 1) $x_{t+1} = x_t - \alpha \cdot 2 \cdot A x_t$
  $= (I - \alpha \cdot 2 \cdot A) x_t$
- ⇒ $V w_{t+1} = (I - \alpha \cdot 2 \cdot V \Lambda V^T) \cdot V w_t$
- ⇒ $V w_{t+1} = V w_t - \alpha \cdot 2 \cdot V \Lambda V w_t$
- ⇒ $w_{t+1} = w_t - \alpha \cdot 2 \cdot \Lambda \cdot w_t$
- ⇒ $w_{t+1,i} = w_{t,i} - \alpha \cdot 2 \cdot \lambda_i \cdot w_{t,i}$; $\forall i$
  $= (1 - \alpha \cdot 2 \cdot \lambda_i) \cdot w_{t,i}$
- ⇒ For convergence: $-1 < 1 - \alpha \cdot 2 \cdot \lambda_i < 1$
  ⇒ $0 < \alpha \cdot 2 \cdot \lambda_i < 2$
  ⇒ $0 < \alpha \cdot \lambda_i < 1$
  ⇒ $0 < \alpha < \frac{1}{\lambda_i}$ $\forall i$

Condition number is $\frac{\lambda_{max}}{\lambda_{min}}$

Stopping Criterion for GD:
- Gradient norm sufficiently small:
  $\|\nabla f(\mathbb{E}[x^t])\|_2^2 \leq \varepsilon$

Benefits of 2nd Order methods
- Gradient might not point directly towards minimum
- Gradient is ignorant of curvature
- Gradient is sensitive to step size-dependent
⇒ 2nd Order methods use inverse of Hessian to scale the gradient
→ use curvature information to estimate direction and size of step

For $f = x^T A x + b^T x + c$ GD and Newton-Raphson are equivalent if $A = \lambda I$ and for GD $\alpha = \frac{1}{\lambda}$

Exercise Sheets

$\frac{S|x-c|^2}{5x} = 2(x-c)^T \cdot \frac{S\|x-c\|_2}{5x} = \frac{(x-c)^T}{\|x-c\|_2}$

$\frac{S u^T v}{5x} = \frac{S u^T}{5x} v + u^T \frac{S v}{5x}$

$\frac{S x^T x}{5x} = \frac{S x^T}{5x} x + x^T \frac{S x}{5x} = \frac{S x^T}{5x} x + x^T \frac{S x}{5x} = 2 x^T \frac{S x}{5x}$

PSD: $A \succeq 0 \Leftrightarrow v^T A v \geq 0 \ \forall v \neq 0$
↳ $v^T (A+B) v = v^T A v + v^T B v$
(pd: is always pos)
A symmetric
⇒ all Eigenvalues of A are ≥ 0
⇒ determinants of submatrices are ≥ 0
(Kriterium Criterion)

negative definite ($\prec 0$) $\det(A_{k \times k}) > 0$
indefinite ⇔ Eigenvalues with opposing signs

Eigenvalues: $\det(A - \lambda I) = 0$
Eigenvector: $(A - \lambda I) \cdot v = 0$
⇒ $A v = \lambda v$

Risk Minimization: $\min \sum_{i=1}^{n} L(y^{(i)}, f(x^{(i)}))$
log. reg: $\min \sum_{i=1}^{n} (0^T x^{(i)} - y^{(i)})^2$

Regularized Risk Minimization: $\min \sum_{i=1}^{n} L(y^{(i)}, f(x^{(i)})) + \lambda \|\theta\|$

directional Slope: $\nabla f(\theta) \cdot u$
↳ maximal for direction of gradient
directional curvature: $u^T H(x) u$
↳ minimal for Eigenvector of largest Eigenvalue

Ridge: $\frac{1}{2} \cdot \frac{\theta^2}{1-\theta^2}$

Digression: → Solving $X^T X \theta = X^T Y$
- 1) Decompose X into Q and R
- 2) Compute $Q^T y$
- 3) Solve triangular System $R\hat{\theta} = Q^T y$ for $\hat{\theta}$
→ Solution because $X^T X \theta = X^T Y \Rightarrow R^T (Q^T Q) R^T \theta = R^T Q^T Y$
$\Rightarrow R^T R \theta = R^T Y$
$\Rightarrow R\theta = Q^T y$

Generality:
- if 2nd derivative available ⇒ use NR
- bei if dimensions get higher ⇒ use GN
- bei if diverging even after ⇒ Adam or GD + mom
  or only 1st derivative

directional derivative:
$D_v f(x) = \nabla f(x) \cdot \hat{v} = \|\nabla f(x)\|_2 \cdot \|\hat{v}\|_2 \cdot \cos(\theta)$

Show Convexity of f
- a) Def: $f(\tau x + (2-\tau)) \leq f(x) + \varepsilon (f(\tau) - f(x))$
- b) 2. derivative ≥ 0 or p.s.d.
- c) sum of two convex functions
- d) Sylvester Criterion (↑)

Standard form in Linearen Programm
- $|x| = r$ wird dargestellt als $v = u - v$; $u, v \geq 0$
  ⇒ $|x| = u + v$
- BeR: $B = B^+ - B^-$; $B^+, B^- \geq 0$
- $ax \leq b_0$ ⇒ $ax + s = b$; $s \geq 0$
⇒ Standard form:
  $\min c^T x$, s.t. $Ax = b$; $x \geq 0$

Taylor 2nd degree
$T_{2,a}(x) = f(a) + \nabla f(a)^T (x-a) + (x-a)^T \nabla^2 f(a)(x-a)$

Ridge Regularized θ:
$\theta^* = (X^T X + \lambda I)^{-1} X^T Y$

↳ objective: $\min_{\theta} \sum_{i=1}^{n} (y_i - \theta_0 - \theta_1 x_i)^2$, s.t. $|\theta|^2 \leq t$

$\pi(z) = \frac{1}{1+\exp(-z)} \Rightarrow \pi'(z) = \pi(z) \cdot (1 - \pi(z))$

$\sigma(\sigma^T x) = \begin{cases} 1 & \sigma^T x \geq 0 \\ 0 & \sigma^T x < 0 \end{cases}$ ist separierbar, weil linke Seite eindeutig gesetzt bei: 0

Hinge Loss: $\max\{1 - y_i \cdot f(x_i), 0\}$
↳ used in a standard SVM with L2-Penalty:
$R(\theta) = \sum_{i=1}^{n} \max\{1 - y_i \cdot f(x_i), 0\} + \lambda \cdot \sum_{j=1}^{d} \theta_j^2$

$0 \leq a_i \leq C$ bei $\beta^T + \beta_0$ „Box constraint"

P, q convex ⇒ $\max\{f, g\}$ convex

$A^T A$ is always p.s.d.
→ $v^T A^T A v = \|Av\|_2^2 \geq 0$

all norms are convex due to triangle inequality